



Fig. 3: A cabinet from Blue Gene/L, a massively parallel processor based supercomputer (Source: www.wikipedia.org)

However, even today, data mining remains more of a standalone activity conducted by a dedicated staff, and its terminology is still quite academically-oriented. This has resulted in data mining as being perceived to be mysterious and somewhat aloof from day-to-day IT projects.

When a business problem is encountered, the IT team starts working on it by developing and understanding data warehouse aspects. The data mining team works differently and contributes by suggesting some kind of analyses or recommendations that will lead to a certain type of action.

Infrastructure requirements

In order to be effective and efficient, the data mining system requires some mandatory inputs. It needs large investments in hardware, software, database, programming communication links and skilled personnel. The volume of data is usually very high in a data mining application. Handling large amounts of data requires software with advanced competencies, and the system needs to be user-friendly and easily accessible by different user departments.



Fig. 4: Crime and Criminal Tracking Network and Systems (CCTNS) has been established to integrate database on crimes by connecting 14,000 police stations in the country (Source: www.skyscrapercity.com)



Fig. 5: Data mining can enable evaluation of the effectiveness of medical treatments (Source: www.atelier.net)

Data warehousing needs to be done carefully as, say, critical marketing actions may be taken based on data analysis. Since data is collected at multiple locations and times, there needs to be a quality control process in place.

Moreover, the database must be continuously updated, as it is a dynamic situation with some old customers leaving, new ones joining and others changing their interests.

Data mining cannot be carried out in isolation. It needs to be integrated with the company's overall marketing strategy.

Finally, data is nothing but numbers. It has to be made to run around, go places and analysed innovatively in order to get good returns on investment.

Currently, data mining applications are offered for all sizes of systems for mainframe, client/server and PC platforms. System prices range from as low as a few thousand dollars for the smallest applications

to as high as a million dollars a terabyte for the largest. Enterprise-wide applications generally range from 10GB to over 10TB.

There are two key technical considerations. First is the size of the database—the larger is the data being processed and maintained, the bigger will be the required system.

Second is query intricacy—the more complex the queries and the larger the number of queries being processed, the more powerful the system required. Relational database storage and management technology is sufficient for most data mining applications less than 50GB. However, this needs to be significantly increased to take on bigger applications.

Some vendors have added widespread indexing capabilities to advance query performance, while others use new hardware architectures such as massively parallel processors (MPPs) to accomplish order-of-magnitude improvements in query time.

Applications

Data mining can be applied to a number of diverse areas and has already proven to be an immensely useful tool.

Law and order. In the last few years there has been a sharp increase in crime rate in India, a country that is as vast as it is diverse. The police have a very challenging role in terms of thinking more intelligently than the criminals and staying ahead of them. One key result area is higher investigative effectiveness and the need for user-friendly interactive data interfaces when looking for clues and verifying records, among other activities.